

NASA SPACE APPS 2026 · CLPS LUNAR MISSION
BROWSER

LunaHorizon

Compare lunar landing sites and dates
in seconds, on the real skyline.

○ Connecting Ridge
● Shackleton crater floor

At the south pole, sunlight and Earth are rationed

1.5°

The Moon's axis tilt. The Sun never climbs more than a couple of degrees, so hills decide who gets light.

55%

Of 2027 that Connecting Ridge, a top Artemis III site, is sunlit. Earth is in view 47%.

0%

Of the year that Shackleton's floor, a few kilometres away, sees the Sun or Earth.

See the Sun and Earth on the real skyline



Real terrain

Horizons traced from NASA LOLA data down to 5 m detail

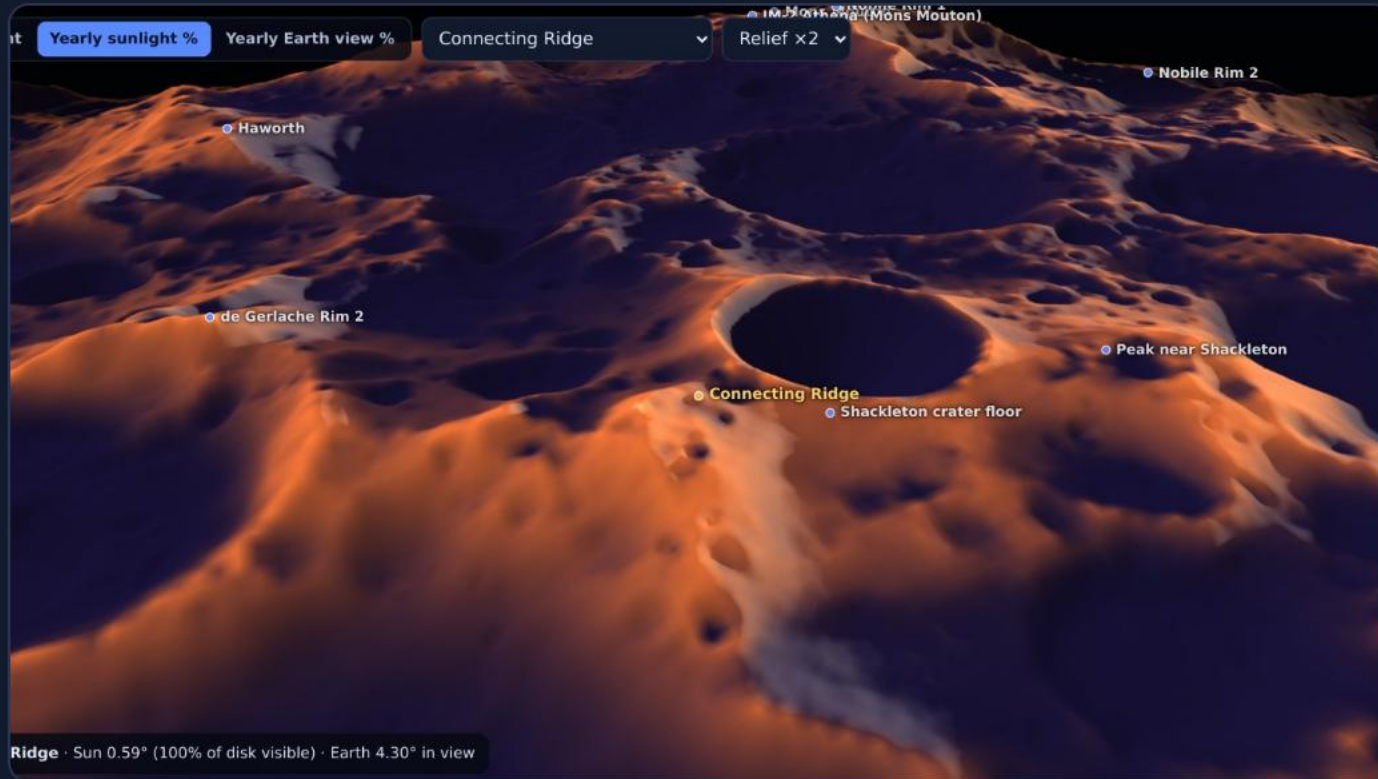
Power and links

Solar watts, battery charge, Deep Space Network and relay coverage

Time machine

Play a lunar day, scrub any date, share the exact view

Every year of the 18.6-year cycle, in 3D



19 years

Sunlight and Earth-view maps for 2026–2044, 160,000 cells each

~3 pts

Typical change in a spot's sunlight between years: the season matters far more

Live

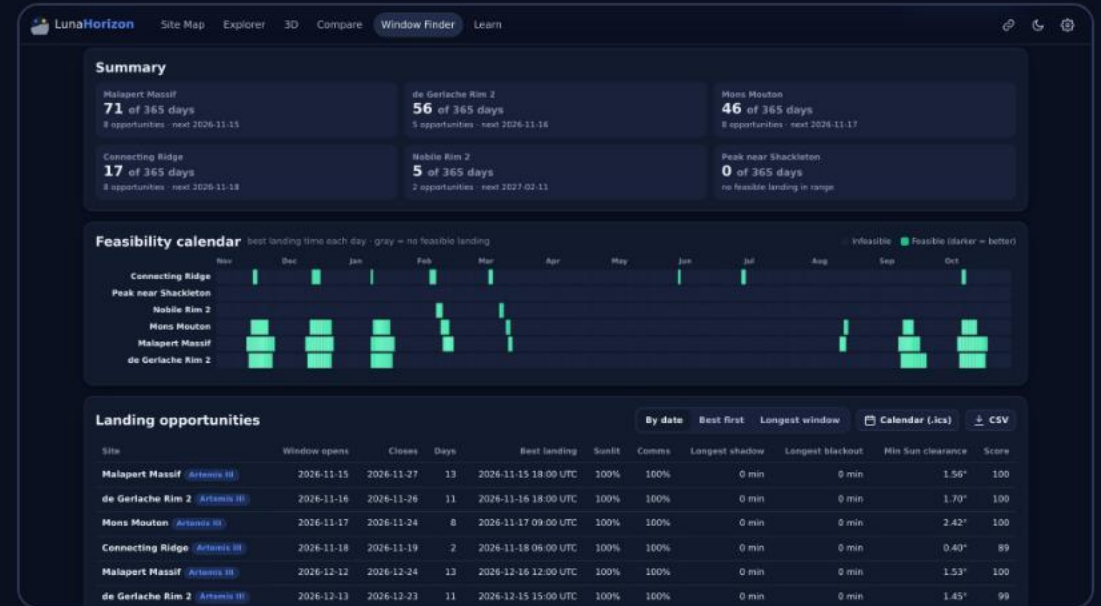
Shadows traced on the GPU toward the Sun, with its true disk size

Which site, which week: answered in seconds



Compare

Rank sites by sunlight, Earth view, comms, longest shadow and energy per day



Window Finder

Every landing time tested against a full Artemis III stay, exported to a calendar

Two choices open new landing windows

With the Sun only a degree or two up, a gentle slope nearby can hide it, and Earth may never rise. LunaHorizon shows what a taller mast or a relay orbiter buys you.

RAISE THE ARRAY

56 → 80%

Malapert Massif sunlit in 2027
with the array at 10 m instead of
2 m

RAISE THE ARRAY

71 → 126

Artemis III landing days at
Malapert in the next 12 months

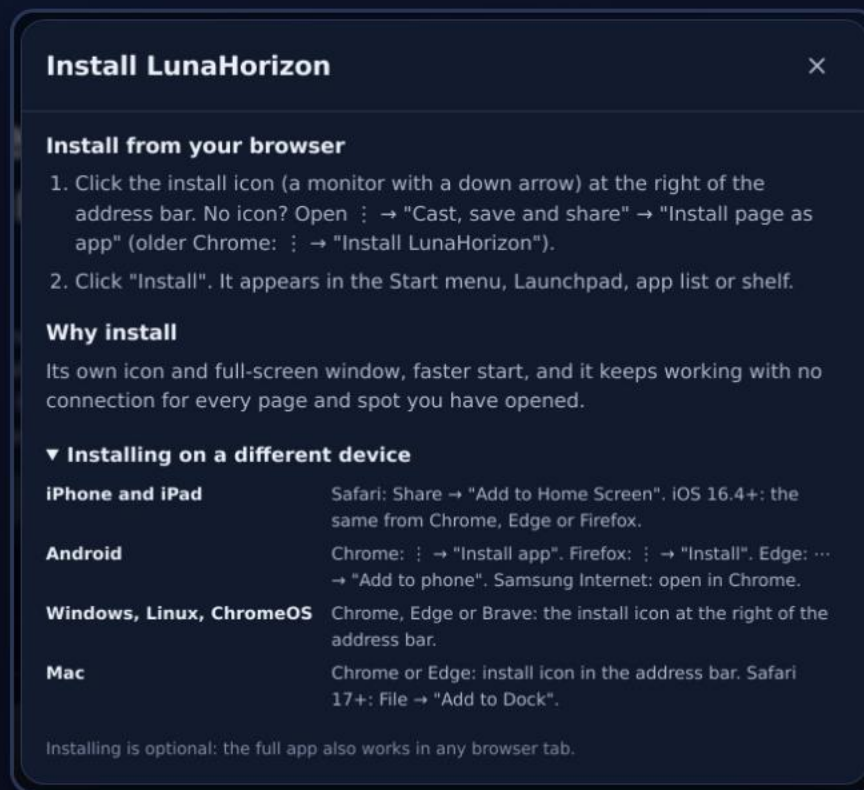
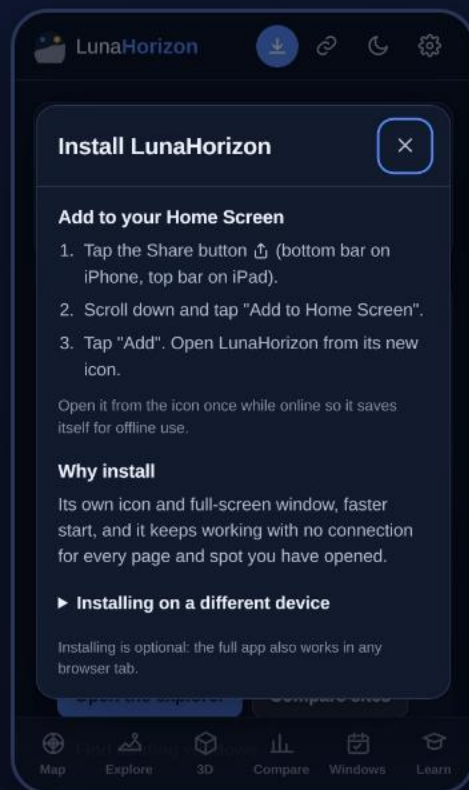
ADD A RELAY ORBITER

0 → 25

CLPS landing days in 3 months at
far-side Schrödinger, which never
sees Earth

Install it on any device, use it offline

One Install button on every phone, tablet and computer, with the exact steps for each browser.



One tap

Where the browser allows it: Chrome, Edge, Samsung, Opera

Every platform

iPhone, iPad, Android, Windows, Mac, Linux, ChromeOS; tested on 20 browser and device setups

Offline

Save everything once (about 12 MB): every page, map year and terrain spot works in airplane mode

Left: iPhone (Safari). Centre: Windows (Chrome). Airplane mode checked on Android and iPhone/iPad.

Built on NASA data, checked against JPL

0.0015°

Mean Sun-position error vs JPL DE421; the Sun's disk is 0.27°

5 m

Finest LOLA terrain in site horizons, out to 260 km

100%

of automated tests pass on every deploy

Offline

Installs on any phone or laptop; no account, no server

TRY IT

samuelakosaonyekwe.github.io/lunahorizon

Code: github.com/samuelakosaonyekwe/lunahorizon